

Project 3

Project Title:

“Customer Churn Prediction & Retention Analysis System”

Objective

To build a Machine Learning system that predicts whether a customer is likely to churn (leave a service) and identifies key factors responsible for churn, enabling businesses to take proactive retention actions.

This project focuses not only on prediction but also on **business insight generation**, which makes it intermediate-level and industry-relevant.

Dataset Description

Use a **Telecom / SaaS Customer Churn Dataset** containing features such as:

- Customer tenure
- Monthly charges
- Total charges
- Contract type (Monthly / Yearly)
- Payment method
- Internet service type
- Customer support interactions
- Add-on services (OTT, security, cloud backup)
- Target variable: **Churn (Yes / No)**

Public datasets like *Telco Customer Churn* can be used.



Detailed Project Workflow

1. Business Understanding & Problem Framing

- Define churn and why it matters to businesses.
- Identify how churn impacts revenue and growth.
- Define success metric (Recall & F1-score over raw accuracy).

2. Data Understanding & Cleaning

- Inspect dataset structure and data types.
- Handle missing values in financial fields.
- Convert TotalCharges-like columns from string to numeric.
- Detect and handle inconsistent or invalid entries.

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3. Feature Engineering (KEY INTERMEDIATE STEP)

Create new meaningful features such as:

- Average revenue per month
- Tenure groups (New / Mid / Loyal customers)
- Service count per customer
- Binary flags for premium service usage
- Contract risk score based on contract type

This step differentiates intermediate projects from basic ones.

4. Exploratory Data Analysis (EDA)

Perform deep analysis:

- Churn rate by contract type
- Churn vs tenure
- Monthly charges distribution for churned vs retained users
- Correlation heatmap
- Churn contribution by services

Use insights to guide modeling decisions.

5. Handling Imbalanced Data

Since churn datasets are often imbalanced:

- Analyze class distribution
 - Apply techniques like:
 - SMOTE
 - Class weights
 - Threshold tuning
 - Justify which method you chose and why.
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6. Model Building

Implement and compare multiple models:

- Logistic Regression (baseline)
- Random Forest
- Gradient Boosting / XGBoost
- Support Vector Machine (optional)

Use pipelines to combine preprocessing and modeling.

7. Model Evaluation & Comparison

Evaluate models using:

- Accuracy (secondary)
- Precision
- Recall (PRIMARY metric)
- F1-Score
- ROC-AUC Curve
- Confusion Matrix



Create a comparison table to justify the final model.

8. Model Interpretation (IMPORTANT)

Explain model decisions using:

- Feature importance (tree-based models)
- SHAP values or permutation importance (optional but recommended)

Explain **why** customers churn, not just **who** churns.

9. Deployment (Optional but Strongly Recommended)

Deploy using:

- Streamlit / Flask / Gradio

UI should include:

- Customer input form
 - Churn probability output
 - Risk category (High / Medium / Low)
 - Key factors influencing prediction
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10. Final Documentation & Business Recommendations

Prepare a final summary including:

- Key churn drivers
 - Retention strategies based on insights
 - Model limitations
 - Future improvement suggestions
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Expected Deliverables

- Clean Jupyter Notebook
- EDA visualizations
- Feature engineering logic
- Trained ML models

- Evaluation comparison table
 - Optional deployed application
 - GitHub repository with README
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Learning Outcomes

By completing this project, the intern will:

- Move from “model building” to “problem solving”
- Learn feature engineering and imbalance handling
- Gain exposure to business-focused ML workflows
- Build an interview-ready, intermediate-level project



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